vision transformer-based sheep breed classification on small datasets for smart farming

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Extended Abstract:

Research area

Accurate sheep breed classification is a modern livestock management which support decisions on breeding, productivity, and profitability in the farming industry. It is a (ViT)-based framework for automatic sheep breed identification, specifically designed to operate effectively even with small datasets. This research contributes to advancing smart agriculture by demonstrating the viability of ViT in automatic sheep breed identification, enabling greater precision and efficiency in farming [1]

Goal/Objectives

The primary objectives are: 1) to enhance sheep breed classification on farms by employing a finetuned ViT trained on a relatively small image dataset 2) to support on productivity,profitability in farms 3)to ensure the enabling farmers,buyers and sellers to clearify breeds in real time.

Methology

The study gives us a structured pipeline to develop an automated sheep breed classification system using ViT. The workflow included dataset preparation, preprocessing, and supervised training of ViT models with systematic hyperparameter tuning.

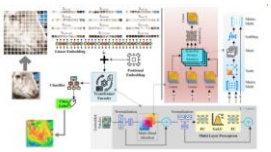


Fig. 1 General architecture of the proposed Vision Transformer model [5]

Result

This study demonstrated the effectiveness of ViT for automatic sheep breed classification, achieving state-of-the-art accuracy of 98.21% and outperforming prior CNN and ensemble-based models on the same dataset. By deploying the trained model into the SheepFormers mobile application, we bridged research and real-world use, providing

Analysis

Traditional CNN architectures such as VGG16 [2] and InceptionV3 [3] achieved accuracies of 95.8% and 92.3%, respectively, while transfer learning with ResNet50V2 [4] reached 97%.

Keywords:*Smart agriculture, Sheep breed classification, Image processing, Vision transformer, Image classification*

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